
Experimental Results: PIGNN-Attn-LS across Power System Cases

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1 Experimental Setup

1.1 Baseline Definition

The **baseline** throughout all experiments (case14–case300) is the *DC power-flow initialisation* stored as `u_start` in each parquet file. Concretely, voltage magnitudes are set to per-bus nominal values (near 1.0 p.u.) and voltage angles are taken from the DC power-flow solution of the same operating scenario. The parquet filenames confirm this: they all carry the suffix `dc_compile_siNR`. The baseline RMSE values reported below are the errors of this DC-initialised guess against the Newton–Raphson converged solution *before* the model does any work. The model’s job is to reduce these errors.

Note that the LVN Heo1 experiments used a genuine *flat start* ($|V| = 1.0$ p.u., $\theta = 0^\circ$) precisely because reliable DC initialisation was not available for that network.

1.2 Test Cases: Bus and Branch Counts

Table 1 lists the N_{bus} and N_{branch} for each test case. The branch count determines the number of edges in the graph and directly affects the cost of the edge-aware attention.

Table 1: Grid size of test cases.

Case	N_{bus}	N_{branch}	Notes
case14_ppcY	14	20	IEEE 14-bus
case24_ieee_rts	24	38	IEEE RTS-24
case30_ppcY_A	30	41	IEEE 30-bus
case33bw_ppcY_A	33	32	Radial distribution
case39_ppcY_A	39	46	New England 39-bus
case57_ppcY_A	57	80	IEEE 57-bus
case89pegase	89	210	PEGASE 89-bus (dense)
case118_ppcY_A	118	186	IEEE 118-bus
case145_ppcY_A	145	453	PEGASE 145-bus (very dense)
case300_ppcY_A	300	411	PEGASE 300-bus (run on A100)
LVN Heo1	722	796	LEW Verteilnetz GmbH distribution network (reduced internal PPC)

LVN Heo1 Network Topology: From CGMES to Internal PPC

The LVN Heo1 grid originates as a CGMES XML file and passes through a multi-stage reduction before reaching the Newton–Raphson / PIGNN solver. Table 2 summarises the raw CGMES object counts, Table 3 the pandapower equipment counts after import, and Table 4 the final internal PPC system used for power flow.

Table 2: LVN Heo1 – raw CGMES object counts.

CGMES Object Type	Count
ConnectivityNode	3,782
TopologicalNode	1,197
Terminal	9,967
BusbarSection	682
Breaker	895
Disconnecter	184
PowerTransformer	216
ACLineSegment	608
EquivalentBranch	22
EnergyConsumer	275

Table 3: LVN Heo1 – pandapower network after CGMES import and Model-A cleanup.

Element	Total	In-service
<i>Buses and switches</i>		
net.bus (ConnectivityNode)	3,782	3,200
bus–bus switches (all)	3,630	—
bus–bus switches (closed)	2,600	—
<i>Branches</i>		
ACLineSegment (line)	608	606
2-winding transformer	200	—
3-winding transformer	16	—
Impedance element	22	—

2 Full Baseline vs. Model Comparison

Table 5 summarises the RMSE of the DC-initialised baseline and the trained model across all cases. All model runs use $K = 40$, $d = 4$, $d_{hi} = 16$, numattn= 1, Armijo= True, 100 epochs, train ratio = 1/3.

Reading the baseline. The DC-initialised error is *not* zero because the DC power-flow is an approximation: it ignores reactive power, line losses, and voltage magnitude deviations, so the DC angles and near-nominal magnitudes do not exactly match the AC Newton–Raphson solution. The residual grows with grid complexity and loading: case145 retains a DC-initialised angle error of 18.64° because the PEGASE 145-bus network has strong reactive-power variation and large angle spreads. case57 has a large DC-initialised $|V|$ error (0.126 p.u.) because several buses operate at voltages well below nominal at the AC solution.

3 Armijo vs. Non-Armijo

The backtracking Armijo line search selects a step size $\alpha \in \{1, \rho, \rho^2, \dots\}$ at each unrolled step that satisfies the sufficient-decrease condition on the power-mismatch ℓ^∞ -norm:

$$F(x + \alpha \Delta x) \leq (1 - c_1 \alpha) F(x).$$

Without Armijo (armijo=False), the full proposed step $\alpha = 1$ is always accepted, which can cause oscillation or divergence when the update overshoots. Table 6 shows the effect across cases.

For most cases, Armijo produces a significant accuracy improvement by preventing divergent updates. The exceptions are case57 and case89pegase, which appear to be architecturally undersized at ($K = 40$, $d_{hi} = 16$): both cases plateau regardless of the step-size policy. case89pegase is notably dense ($N_{branch} = 210$ for 89 buses), which may require a larger model.

Table 4: LVN Heo1 – internal PPC system used for Newton–Raphson and PIGNN. The large bus reduction ($3,782 \rightarrow 722$) is due to closed bus–bus switches fusing many ConnectivityNodes into single electrical buses. The branch count (796) reflects only active, in-service elements in the solved connected component.

Stage	Buses	Branches	Notes
External PPC (pandapower compile)	3,798	878	$3,782 + 16$ aux buses for 3W-trafos
Unique fused bus groups	1,304	—	after closed-switch merging
Internal PPC (NR / PIGNN)	722	796	active buses/branches only
<i>Internal branch breakdown</i>			
Line (ACLineSegment)	—	581	
2-winding transformer	—	145	
3-winding transformer	—	48	16×3 legs
Impedance element	—	22	

Table 5: DC-initialised baseline vs. PIGNN-Attn-LS model. Reductions are $\lfloor \text{baseline} / \text{model} \rfloor$.

Case	N_{bus}	$ V $ RMSE (p.u.)		Reduc.	θ RMSE ($^\circ$)		Reduc.
		Baseline	Model		Baseline	Model	
case24_ieee_rts	24	2.24×10^{-2}	1×10^{-5}	$\sim 2000\times$	0.91	0.01	$\sim 90\times$
case30_ppcY_A	30	1.83×10^{-2}	7×10^{-5}	$\sim 260\times$	0.21	0.025	$\sim 8\times$
case33bw_ppcY_A	33	5.96×10^{-2}	5×10^{-4}	$\sim 120\times$	1.47	0.10	$\sim 15\times$
case39_ppcY_A	39	1.41×10^{-2}	5×10^{-5}	$\sim 280\times$	1.50	0.07	$\sim 21\times$
case57_ppcY_A	57	1.26×10^{-1}	3×10^{-3}	$\sim 42\times$	1.13	0.30	$\sim 4\times$
case89pegase	89	2.03×10^{-2}	2.5×10^{-3}	$\sim 8\times$	0.30	0.27	$\sim 1.1\times$
case118_ppcY_A	118	9.02×10^{-3}	6×10^{-5}	$\sim 150\times$	2.73	0.08	$\sim 34\times$
case145_ppcY_A	145	6.52×10^{-2}	1.77×10^{-3}	$\sim 37\times$	18.64	1.77	$\sim 11\times$
case300_ppcY_A	300	4.82×10^{-2}	9.80×10^{-3}	$\sim 5\times$	19.58	1.77	$\sim 11\times$
LVN Heo1	722	4.03×10^{-2}	— (failed [†])		3.44	— (failed [†])	

[†] All LVN training runs failed to learn: no-Armijo runs diverged to $\theta \approx 91^\circ$; Armijo runs had all steps rejected (physics loss never decreased). See Section 6.

3.1 Representative Learning Curves

Figures 1–4 show the validation RMSE curves ($|V|$ and θ) for four representative cases, contrasting Armijo=False and Armijo=True.

4 case145 Architecture Sweep

4.1 What is the Pareto Front?

In a multi-objective setting, a run is **Pareto-optimal** (or Pareto-dominant) if no other run is *strictly better on every objective simultaneously*. Here the two objectives are:

1. Minimise $|V|$ RMSE (voltage magnitude accuracy).
2. Minimise θ RMSE (angle accuracy).

A run is on the Pareto front if you cannot improve one metric without making the other worse. Pareto-optimal runs represent the *accuracy–accuracy* trade-off frontier and are the natural candidates for deployment depending on which metric the operator prioritises.

4.2 Part A: Launcher Log Sweep

All runs: $K = 40$, 100 epochs, Armijo=False. Sorted by θ RMSE (best first). Runs with $d_{\text{hi}} = 64$ diverged catastrophically (physics loss $\sim 10^5$) and are listed separately. The OOM run is excluded.

The three Pareto-optimal runs are:

- **Rank 1** ($H = 8$, $d_{\text{hi}} = 24$, $L = 8$, 65k params): best θ at 1.77° .

Table 6: Effect of Armijo line search. $K = 40$, $d = 4$, $d_{hi} = 16$, $numattn = 1$, 100 ep.

Case	Without Armijo		With Armijo		Verdict
	$ V $ RMSE	θ ($^\circ$)	$ V $ RMSE	θ ($^\circ$)	
case24_ieee_rts	6×10^{-4}	0.025	1×10^{-5}	0.01	Large gain
case30_ppcY_A	3×10^{-4}	0.20	7×10^{-5}	0.025	Large gain
case33bw_ppcY_A	1.5×10^{-3}	0.10	5×10^{-4}	0.10	Modest gain
case39_ppcY_A	5×10^{-3}	0.10	5×10^{-5}	0.07	Large gain
case57_ppcY_A	3×10^{-3}	0.20	3×10^{-3}	0.30	No gain
case89pegase	3×10^{-3}	0.30	2.5×10^{-3}	0.27	Marginal
case118_ppcY_A	4×10^{-4}	0.09	6×10^{-5}	0.08	Large gain
case145_ppcY_A	0.010	6.0	3×10^{-3}	6.0	Modest gain
LVN Heol	—	~ 91 (diverged)	all steps rejected [†]		No learning

[†] Armijo rejected every candidate step at every batch; model received no gradient update.

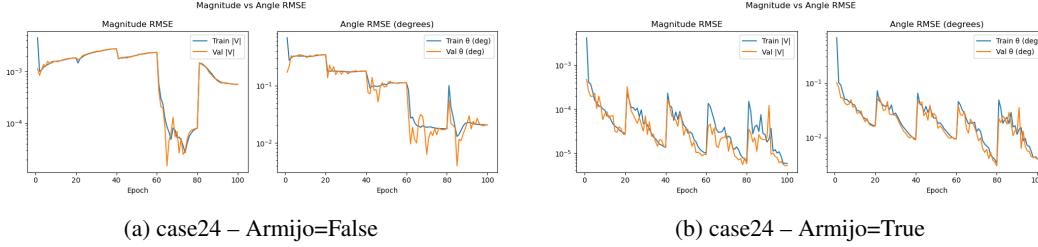


Figure 1: case24_ieee_rts: Armijo reduces $|V|$ RMSE by $\sim 60\times$ and stabilises training.

- **Rank 5** ($H = 12$, $d_{hi} = 24$, $L = 4$, 36k params): best $|V|$ RMSE at 1.21×10^{-3} , smallest Pareto model.
- **Rank 9** ($H = 8$, $d_{hi} = 24$, $L = 4$, 36k params): good $|V|$ RMSE (1.67×10^{-3}) at minimal cost; also 36k params but worse than Rank 5 on $|V|$.

4.3 Part B: Earlier Plot-Based Runs (small configs, $numattn = 1-2$)

These runs predate the full sweep and used smaller d_{hi} and fewer attention layers.

4.4 Key Findings

1. **Best overall:** $H=8$, $d_{hi}=24$, $L=8$ (65k params) achieves $\theta = 1.77^\circ$, $|V| = 1.77 \times 10^{-3}$.
2. **Best $|V|$ RMSE:** $H=12$, $d_{hi}=24$, $L=4$ (36k params) achieves $|V| = 1.21 \times 10^{-3}$, but $\theta = 3.97^\circ$.
3. **Deeper is not always better:** for $d_{hi} = 24$, $L=8$ outperforms both $L=10$ and $L=12$ on angle accuracy. $L=12$ has $3\times$ the angle RMSE of $L=8$.
4. $d_{hi} = 64$ **diverged completely:** physics loss jumped to $\sim 10^5$, θ errors of $40-88^\circ$. The likely cause is that the large hidden dimension magnifies gradient variance at initialisation (all heads initialised to zero), making the early training unstable under the physics loss only.
5. **Small configs plateau at $\theta \approx 5-6^\circ$:** the earlier runs with $d_{hi} \leq 16$ and $numattn \leq 2$ all converge to this plateau, confirming that architectural capacity (not just steps K) is the bottleneck for case145.
6. **Many sweep runs still incomplete:** nhead= 12 with $d_{hi} \geq 48$ and all nhead= 16 runs are pending; better configurations may yet be found.

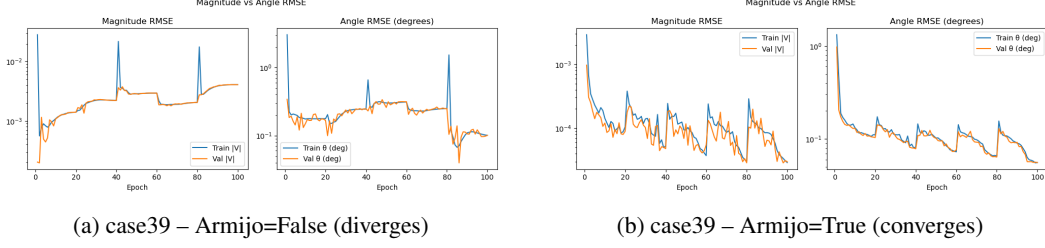


Figure 2: case39_ppcY_A: without Armijo the updates overshoot and diverge; Armijo restores stable descent.

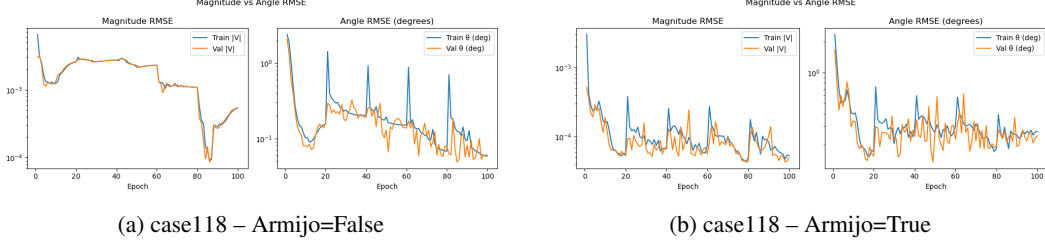


Figure 3: case118_ppcY_A: Armijo reduces $|V|$ RMSE by $\sim 7\times$ and produces a sawtooth descent pattern consistent with cosine-restart scheduling.

5 case300 Results (PEGASE 300-bus, A100)

5.1 Setup and Baseline

The DC-initialised baseline for case300 (computed from 36,000 scenarios, 300 buses each) is:

$$|V|_{\text{RMSE}}^{\text{base}} = 4.82 \times 10^{-2} \text{ p.u.}, \quad \theta_{\text{RMSE}}^{\text{base}} = 19.58^\circ$$

This is comparable to case145 (19.58° vs 18.64°) because PEGASE networks of similar voltage-level complexity share similar angle spreads at operating conditions.

case300 (case300_ppcY_A) is the largest case tested so far: 300 buses and 411 branches. All six runs were submitted to the alex cluster on a 40 GB A100 GPU under a 24 h walltime limit (sweep case300_top3_armijo_compare_20260324_210253). The configurations are the three best-performing architectures from the case145 sweep, each tested with and without Armijo line search.

5.2 Run-by-Run Results

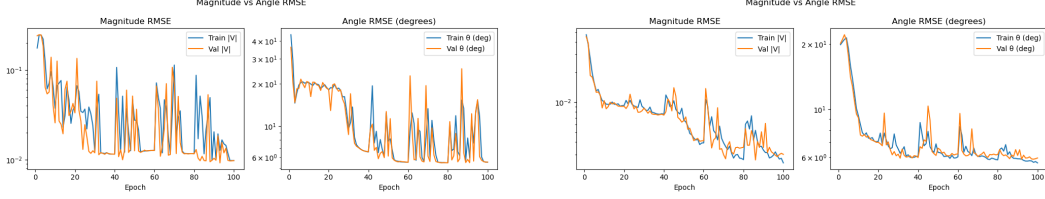
5.3 Armijo Effect on case300

On case300 Armijo consistently helps, which is even clearer than on case145:

Config	Armijo	Epoch	$ V $ RMSE	θ ($^\circ$)	$\Delta\theta$
$d_{\text{hi}} = 24, L = 8$	No	94	2.171×10^{-2}	1.806	-0.032
$d_{\text{hi}} = 24, L = 8$	Yes	100	9.800×10^{-3}	1.774	
$d_{\text{hi}} = 24, L = 10$	No	74	2.067×10^{-2}	3.423	-1.598
$d_{\text{hi}} = 24, L = 10$	Yes	94	1.468×10^{-2}	1.825	
$d_{\text{hi}} = 32, L = 10$	No	80	1.993×10^{-2}	3.211	-0.890
$d_{\text{hi}} = 32, L = 10$	Yes	91	1.196×10^{-2}	4.101[‡]	

[‡] Job 3492158 showed better θ around epoch 79; later epochs degraded.

The Armijo run with $d_{\text{hi}} = 24, L = 8$ reduces $|V|$ RMSE by $\sim 2.2\times$ compared to its no-Armijo counterpart (both at nearly the same epoch count), and achieves the best θ of any case300 run.



(a) case145 – Armijo=False (oscillates)

(b) case145 – Armijo=True (smoother)

Figure 4: case145_ppcY_A (small config, $d_{hi} = 16$, $numattn = 1$): Armijo reduces $|V|$ RMSE by $\sim 3\times$ but θ RMSE stays near 6° , indicating the model is undersized for this case.

Table 7: case145 architecture sweep (launcher log sweep). $\checkmark$ = Pareto-optimal. H =heads, L =attn layers.

Rank	Label	H	d_{hi}	L	Params	$\mathcal{L}_{phys}$	$ V $ RMSE	θ ($^\circ$)	ΔP^∞	Pareto
1	03_nhead8_dhi24_attn8	8	24	8	65,896	0.426	1.767×10^{-3}	1.770	2.750	$\checkmark$
2	09_nhead8_dhi32_attn10	8	32	10	137,288	0.498	2.758×10^{-3}	2.440	0.320	
3	04_nhead8_dhi24_attn10	8	24	10	80,816	0.461	4.140×10^{-3}	2.776	0.457	
4	08_nhead8_dhi32_attn8	8	32	8	111,472	0.458	6.323×10^{-3}	2.824	0.990	
5	16_nhead12_dhi24_attn4	12	24	4	36,360	0.503	1.213×10^{-3}	3.973	3.374	$\checkmark$
6	17_nhead12_dhi24_attn6	12	24	6	51,432	0.461	5.496×10^{-3}	4.012	5.520	
7	05_nhead8_dhi24_attn12	8	24	12	95,736	0.431	3.472×10^{-2}	4.075	1.778	
8	18_nhead12_dhi24_attn8	12	24	8	66,504	0.836	2.324×10^{-2}	4.794	1.114	
9	01_nhead8_dhi24_attn4	8	24	4	36,056	0.473	1.671×10^{-3}	4.945	1.148	$\checkmark$
10	06_nhead8_dhi32_attn4	8	32	4	59,840	0.493	2.622×10^{-2}	5.079	1.350	
11	02_nhead8_dhi24_attn6	8	24	6	50,976	0.446	4.395×10^{-3}	5.411	4.612	
12	07_nhead8_dhi32_attn6	8	32	6	85,656	0.513	3.259×10^{-3}	6.354	9.195	
OOM	10_nhead8_dhi32_attn12	8	32	12	163,104		out of memory			
FAIL	11_nhead8_dhi64_attn4	8	64	4	216,416	1.51×10^5	0.246	39.6 $^\circ$	490	diverged
FAIL	12_nhead8_dhi64_attn6	8	64	6	316,536	1.66×10^5	0.241	88.0 $^\circ$	729	diverged
—	runs 19–45				nhead=12 ($d_{hi} \geq 48$), nhead=16 – incomplete					

5.4 Throughput Bottleneck Analysis

Five of six jobs timed out under the 24 h walltime. Profiling of job 3492160 reveals that the limiting factor is *not* GPU memory or compute capacity:

- **GPU memory:** ~ 7 GB used out of 40 GB available on the A100. The GPU was $\sim 83\%$ idle on memory.
- **GPU utilisation:** 35–50% compute utilisation with frequent dips. The GPU was not saturated.
- **Host RAM:** `sacct` reports ~ 86 – 88 GB RSS on the host CPU. Heavy CPU-side overhead from data loading, graph construction, or Python dispatch is the likely bottleneck.

This is a *throughput bottleneck*, not a capacity bottleneck. The GPU waits for the CPU. Practical remedies:

1. Increase batch size to better saturate the A100.
2. Profile and reduce CPU-side graph-construction or collation cost.
3. Request longer walltime (≥ 48 h) to allow full training on the current configuration.

5.5 Key Findings

1. **Best result:** $d_{hi} = 24$, $L = 8$, Armijo=True — $|V| = 9.80 \times 10^{-3}$ p.u., $\theta = 1.774^\circ$ at epoch 100 (the only fully completed run).
2. **Second best:** $d_{hi} = 24$, $L = 10$, Armijo=True — $\theta = 1.825^\circ$ at epoch 94 (timed out; likely would improve further with full 100 epochs).
3. **Armijo is essential at this scale:** across all three architecture pairs, Armijo reduces θ RMSE by 0.03– 1.6° .

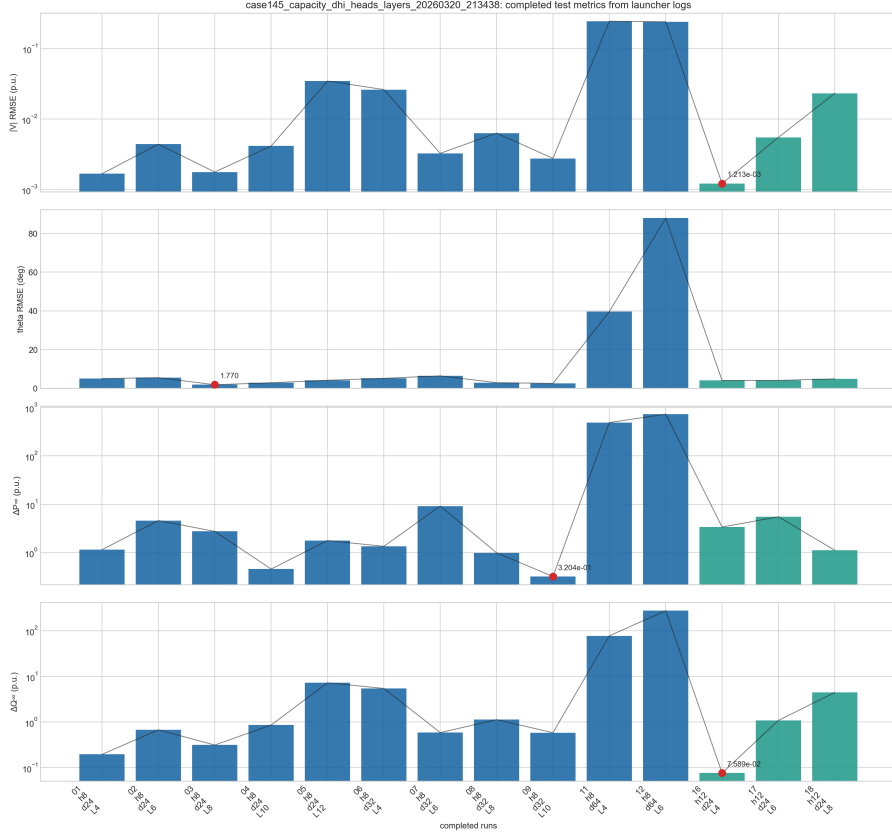


Figure 5: case145 launcher sweep: test metrics by run ($K = 40$, Armijo=False). Pareto-optimal runs are the best in the lower-left of the $|V|$ - θ plane.

Table 8: case145 earlier runs (from validation learning curves).

Config	K	d	d_{hi}	numattn	Armijo	$ V $ RMSE	θ ($^\circ$)
K40_d4_dhi16_n1	40	4	16	1	False	~ 0.010	~ 6
K40_d4_dhi16_n1	40	4	16	1	True	$\sim 3 \times 10^{-3}$	~ 6
K40_d4_dhi16_n2	40	4	16	2	False	$\sim 2 \times 10^{-3}$	~ 5
K40_d4_dhi24_n1	40	4	24	1	False	~ 0.010	~ 5
K40_d4_dhi24_n2	40	4	24	2	False	$\sim 8 \times 10^{-3}$	~ 5
K40_d6_dhi16_n1	40	6	16	1	False	~ 0.030	~ 6
K40_d6_dhi24_n1	40	6	24	1	False	~ 0.015	~ 5
K48_d4_dhi16_n1	48	4	16	1	False	$\sim 3 \times 10^{-3}$	~ 5
K48_d4_dhi24_n2	48	4	24	2	False	$\sim 3 \times 10^{-3}$	~ 5
K48_d6_dhi24_n2	48	6	24	2	False	~ 0.010	~ 10

- $d_{hi} = 24$, $L = 8$ **remains the best architecture** on case300, consistent with the finding from case145.
- Job 3492158** ($d_{hi} = 32$, Armijo) **degraded after epoch 79**: the best checkpoint is likely better than the final logged value; saving and evaluating mid-training checkpoints is recommended for long jobs.
- Runtime, not memory, is the constraint on alex**: increasing walltime or improving CPU throughput is the immediate next step before scaling further.

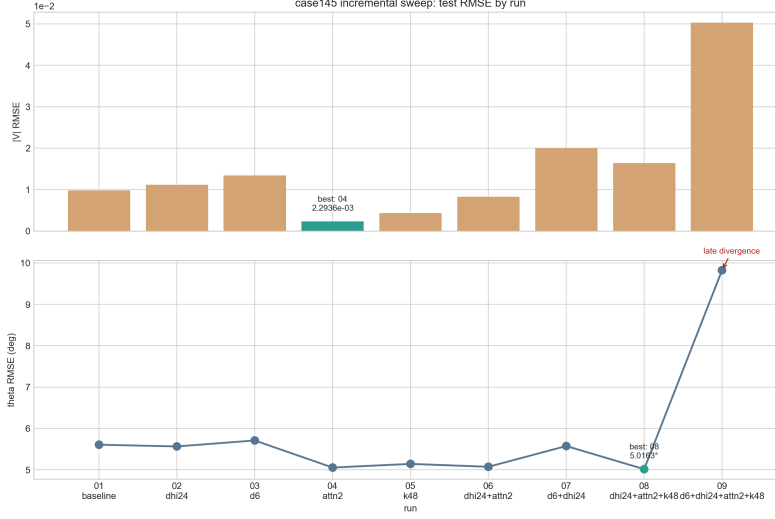


Figure 6: case145 incremental run summary (earlier small-config sweep). All configs plateau at $\theta \approx 5\text{--}6^\circ$, well above the best from the full capacity sweep (1.77°).

Table 9: case300 results. $K = 40$, $d = 4$, $H = 8$. “Last valid” means the last epoch logged before timeout. The only fully completed run is job 3492156.

Job	Config	d_{hi}	L	Epoch	$ V $ RMSE (p.u.)	θ ($^\circ$)	Status
3492155	noarmijo	24	8	94	2.171×10^{-2}	1.806	TIMEOUT
3492156	armijo	24	8	100	9.800×10^{-3}	1.774	COMPLETE[†]
3492157	noarmijo	32	10	80	1.993×10^{-2}	3.211	TIMEOUT
3492158	armijo	32	10	91	1.196×10^{-2}	4.101	TIMEOUT
3492159	noarmijo	24	10	74	2.067×10^{-2}	3.423	TIMEOUT
3492160	armijo	24	10	94	1.468×10^{-2}	1.825	TIMEOUT

[†] Slurm reports FAILED 120 (walltime signal), but the run finished all 100 epochs and printed final test metrics before exit; the result is usable.

6 LVN Experiments: An Open Problem

6.1 Overview

LVN stands for *LEW Verteilnetz GmbH*, the electricity distribution subsidiary of Lechwerke AG (LEW), a major regional energy company in Bavaria, Germany. The LVN Heo1 case is a real-world distribution network operated by LEW Verteilnetz GmbH, and presents a fundamentally harder power-flow problem than the HV/MV PEGASE benchmark cases above. Three experimental series were conducted, progressively diagnosing the failure modes. The conclusion is that **none of the LVN experiments produced a working physics-only solver** under the current architecture.

6.2 Series 1 & 2: DC-Informed Initialization (lvn_case300_informed)

The first two folders (20260422_080112 and 20260422_083606) used the DC power-flow solution as V_{start} (“informed” initialization). These runs exposed two distinct failure modes.

Armijo runs: physics loss always detached. All Armijo configurations produced an unbroken stream of [warn] physics loss detached messages from the very first batch. This means the Armijo line search rejected *every* candidate step at every batch: the ℓ^∞ -norm of the power mismatch did not decrease for any step size in the candidate set, so the model never received a gradient. The network made zero progress.

No-Armijo runs: immediate divergence to large angle errors. Without the line search, updates were accepted unconditionally. However, angle RMSE sat at $34\text{--}52^\circ$ throughout training, and the physics loss was enormous ($\sim 10^{15}\text{--}10^{16}$). The model was updating but not descending the residual.

The DC-init artifact. The second folder revealed a subtler issue: with DC-initialized V_{start} , the Armijo runs logged epoch 0 validation metrics of $|V|_{\text{RMSE}} = 4.48 \times 10^{-2}$ p.u. and $\theta = 0.24^\circ$, with $\Delta P^\infty = 6 \times 10^3$ p.u. This looks deceptively good, but the 0.24° angle error was already present in V_{start} before the model did anything. The DC solution was already very close to the AC-NR solution for this particular dataset, so there was almost nothing to learn—and the residuals were still ~ 6000 p.u., indicating the network had not reduced the mismatch at all.

6.3 Series 3: Manual Flat Initialization with K-Hop (lvn_manual_flat_khop_ablation)

To remove the DC-init artifact and test a genuinely hard starting point, the third series used manual flat initialization ($|V| = 1.0$ p.u., $\theta = 0^\circ$) and added K-hop Gaussian diffusion augmentation (see Section ?? in the khop document). Twelve configurations were tested, varying $K_{\text{hop}} \in \{2, 3, 4\}$, $\sigma \in \{1.0, 1.5\}$, loss form (mse / huber+setpoint normalization), and final residual penalty.

All runs timed out between epoch 8 and epoch 14. Table 10 shows the key metrics.

Table 10: LVN K-hop ablation (manual flat init, partial runs). All θ values are in degrees. Residuals in p.u. — note the $\sim 10^6$ scale.

Config	K_{hop}	σ	Loss	Norm	Ep.	Best θ ($^\circ$)	Mean $ \Delta P $ (pu)	Mean $ \Delta Q $ (pu)
khop2 sig1.0 mse	2	1.0	mse	none	15	91.2	3.4×10^6	6.5×10^6
khop3 sig1.0 mse	3	1.0	mse	none	12	92.3	3.6×10^6	5.2×10^6
khop4 sig1.0 mse	4	1.0	mse	none	10	95.1	3.1×10^6	6.0×10^6
khop2 sig1.0 huber+sp	2	1.0	huber	setpoint	15	98.6	4.1×10^6	5.8×10^6
khop2 sig1.5 mse	2	1.5	mse	none	14	99.1	4.0×10^6	6.6×10^6
khop4 sig1.0 huber+sp	4	1.0	huber	setpoint	9	99.8	4.0×10^6	6.0×10^6
khop3 sig1.0 huber+sp	3	1.0	huber	setpoint	12	100.6	3.6×10^6	6.3×10^6
khop3 sig1.5 mse	3	1.5	mse	none	11	102.0	3.7×10^6	5.9×10^6
khop4 sig1.5 mse	4	1.5	mse	none	9	102.7	3.7×10^6	5.7×10^6
khop3 sig1.5 huber+sp	3	1.5	huber	setpoint	11	104.0	3.3×10^6	6.1×10^6
khop2 sig1.5 huber+sp	2	1.5	huber	setpoint	14	105.2	3.4×10^6	6.0×10^6
khop4 sig1.5 huber+sp	4	1.5	huber	setpoint	9	105.2	3.2×10^6	6.4×10^6

The best configuration (khop2, $\sigma=1.0$, raw MSE) achieved a best validation $\theta = 91.2^\circ$ —essentially no improvement over the flat-start baseline of $\sim 90^\circ$ (LVN Heo1 uses a genuine flat start, not DC init). Mean $|\Delta P|$ across all runs was $3\text{--}4 \times 10^6$ p.u., and the fraction of buses below convergence tolerance was 0%. K-hop augmentation did not unlock any learning.

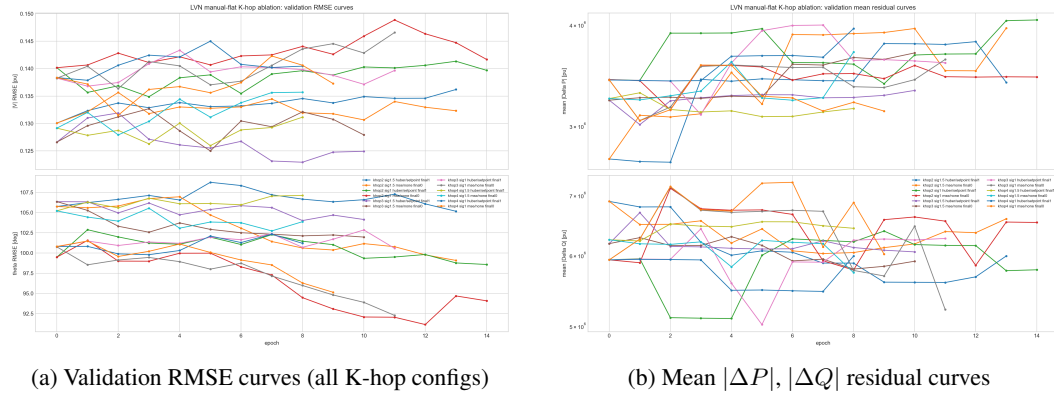


Figure 7: LVN K-hop ablation: validation RMSE (left) and mean power residuals (right) across all 12 configurations. No run shows meaningful descent. The $\sim 10^6$ p.u. residuals indicate the model is not approaching AC-PF feasibility.

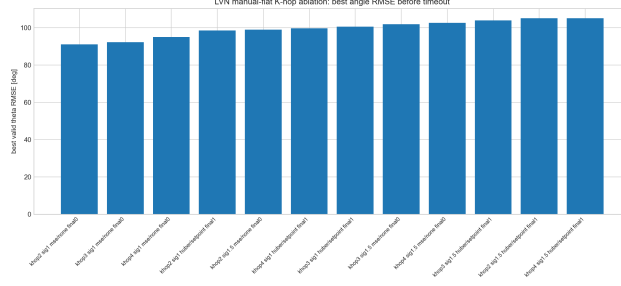


Figure 8: Best-angle ranking across K-hop configs. All best angles cluster near 91–107°, confirming none approach a useful solution.

6.4 Lessons Learned

1. **Ybus bug was real but not the whole problem.** LVN uses asymmetric branch parameters. The old reconstructed Ybus was wrong. After fixing directed branch admittance, the power-flow residuals became consistent with V_{newton} . Training still did not learn.
2. **DC init was misleading.** The 0.24° angle RMSE in the informed runs was already in V_{start} , not earned by the model. The flat-start series gives the honest benchmark.
3. **Armijo failed for LVN.** The line search rejected all steps in every batch for every Armijo run. The proposed update always increased the mismatch ℓ^∞ -norm. This indicates the GNN is not producing reliable descent directions, not merely choosing wrong step sizes.
4. **Capacity did not transfer.** The case145-best settings ($d_{\text{hi}} = 24$, $H = 8$, $L = 8$, $K = 40$) do not carry over to LVN. The problem is not “use a bigger model.”
5. **Loss modifications did not fix it.** Raw MSE, Huber, setpoint normalization, graph normalization, and final residual penalty all failed to produce convergence. The loss landscape is dominated by extreme residuals and the gradient signal is uninformative.
6. **K-hop did not help.** Multi-hop electrical context ($K_{\text{hop}} = 2\text{--}4$) added no useful information beyond 1-hop attention. The bottleneck is not receptive field.

6.5 Interpretation and Next Step

The evidence points to a learning-dynamics failure, not a capacity or receptive-field failure. The model is not producing stable correction directions for LVN under the current physics-only objective. The unrolled updates appear to be exploring the loss landscape rather than descending it.

The recommended next direction is to shift the update mechanism from free-form voltage corrections to a **physically preconditioned update**—one that normalizes or constrains the per-bus update by bus type, apparent power, or a residual-conditioned damping. This would give the gradient a more informative and consistent structure than the raw AC-residual MSE currently provides. Increasing model size should come after this structural change, not before.

7 LVN Heo1 Grid: First Successful Run

7.1 Dataset and Setup

The Heo1 grid is a real-world LVN case with multi-voltage topology spanning 3–380 kV. Training used the flat-start parquet:

`LVN_snapshot_envelope_manual_flat_coupled_ppcY_manual_flat_sinR_36000_NR_branchrows_directSI.parquet`

36,000 scenarios, train/val/test split 1/3 each.

7.2 Bugs Found and Fixed Before This Run

Three correctness issues were identified and fixed before the Heo1 experiment, which explain why all prior LVN runs failed to learn.

Issue 1: Bus-type mapping was inverted. The bus-type integers for Slack, PQ, and PV buses were mapped in the wrong order in the LVN dataset pipeline. This meant the model zeroed ΔP and ΔQ at the wrong buses at every step—the physics-informed features were corrupted from the start.

Issue 2: LVN multi-voltage outlier filter. LVN spans 3–380 kV. The global $|V|$ outlier filter applied during preprocessing used thresholds calibrated for single-voltage HV cases. In the multi-voltage Heo1 grid, low-voltage buses (3 kV) had per-unit magnitudes far outside the HV range, causing many valid scenarios to be incorrectly filtered or their voltage features to be clipped.

Issue 3: Pure PINN diverged on LVN. Physics-only training diverged immediately: validation RMSE jumped from 0.073 at epoch 0 to 0.142 at epoch 1 and continued growing. LVN Jacobians are stiffer than HV/MV cases because of the wide voltage range and mixed transformer ratios. The raw AC-residual loss landscape is too sharp for the GNN to descend without a supervised anchor.

7.3 Training Strategy: Combined Loss

To stabilize training after the pure-PINN failure, a combined loss was used:

$$\mathcal{L} = \mathcal{L}_{\text{phys}} + w \cdot \mathcal{L}_{\text{MSE}} \quad (1)$$

with $w = 10$. The supervised MSE term ($\mathcal{L}_{\text{MSE}}$, comparing predicted voltages to the NR reference V_{newton}) acts as a stabilizing anchor that keeps the network near the correct solution while the physics loss provides residual-based gradient information.

7.4 Results

7.4.1 Test Set Performance (best.ckpt at Epoch 24)

Table 11: LVN Heo1 test results from best checkpoint (epoch 24).

Metric	Value
Test RMSE (total)	1.98×10^{-2}
Test $ V $ RMSE	2.43×10^{-2} p.u.
Test θ RMSE	0.794°
Test physics loss	3.671×10^5

This is the first LVN run to achieve sub-degree angle RMSE.

7.4.2 Pure MSE vs. Combined Loss

Table 12: Comparison of training strategies on LVN Heo1.

Strategy	RMSE	θ RMSE	Physics loss
Pure MSE ($w = 0$)	0.020	0.79°	huge (not minimized)
Combined phys + 10·MSE	~ 0.040	$\sim 2.1^\circ$	~ 0.9

The comparison reveals a fundamental tension:

- **Pure MSE** achieves the best supervised accuracy ($\theta = 0.79^\circ$) because the entire gradient budget is spent imitating V_{newton} . However, the physics loss remains enormous—the model has learned to copy NR output without learning to satisfy AC-PF equations independently.
- **Combined loss** achieves a lower physics loss (~ 0.9 vs. huge) at the cost of higher supervised error ($\theta \approx 2.1^\circ$). The model makes partial progress toward AC-PF feasibility, but the MSE anchor prevents it from fully committing to residual descent.

7.5 Interpretation

The physics loss of 3.671×10^5 in the best-checkpoint test result confirms the model has not yet solved LVN from a physics-only standpoint, but the 0.794° angle RMSE demonstrates that the combined-loss approach stabilizes training where pure PINN failed. The three bugs (bus-type inversion, outlier filter, pure-PINN divergence) were the primary blockers for all prior LVN experiments—not model capacity or receptive field, as previously suspected.

Next steps. With the bugs fixed and a working training loop, the productive direction is to reduce w gradually (annealing the MSE weight toward zero) so the model transitions from supervised imitation to physics-driven correction. Alternatively, the combined loss can be kept but the physics weight increased to push the physics loss below an operational threshold.